Incorporating RAG for Factual Hallucination Detection Modeling in Intelligent Educational system

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2025 7th International Conference on Computer Science and Technologies in Education (CSTE) | 979-8-3315-1166-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/CSTE64638.2025.11092000

***Abstract*—Factual hallucination is one of the key factors affecting learning effectiveness and knowledge construction in intelligent education system, and reducing factual hallucination can help develop learners' critical thinking and accurate knowledge acquisition. How to effectively detect factual hallucination in intelligent education system is a challenging issue in educational technology research. Based on this, this study proposed a factual hallucination detection model that incorporates knowledge retrieval enhancement and fine-tuning of large language model. First, an educational knowledge vector base is constructed from the collected knowledge datasets in the educational domain; then, a factual illusion detection model is constructed by fusing the retrieval augmented generation(RAG) technique and Low-Rank Adaptation of Large Language Models(LoRA) fine-tuning method to optimize the large language model. We conducted experiments on a real dataset in the education domain, and the results showed that this experimental model outperforms the existing models in terms of precision, recall, and F1 score for factual hallucination detection based on knowledge retrieval enhancement and model fine-tuning. Therefore, the experimental model in this paper is able to effectively detect factitious hallucination in intelligent education system, which lays the foundation for providing high-quality educational content.**

Keywords—intelligent educational system, factual hallucination, RAG

I. Introduction

Intelligent education system, as an important outcome of education information, are providing learners with intelligent and personalized education services by integrating advanced technologies [1] . These system play an important role in supporting distance learning and guaranteeing educational equity, showing great application value and development potential [2] . However, with the wide application of big language modeling in these system, the problem of factitious illusion of model output is becoming more and more prominent and has become a challenge that needs to be solved urgently. Factual hallucination refer to the possibility that the model may generate smoothly expressed but factually incorrect content, e.g., making up non-existent concepts, confusing historical facts, or giving false scientific explanations [3] . In educational scenarios, factual hallucination may lead to the dissemination of erroneous knowledge, reduce system credibility, and pose potential risks in certain specialized areas, thus seriously affecting learning outcomes and cognitive development [4] . Therefore, the establishment of an accurate and reliable detection mechanism for factitious hallucination has become a critical need to safeguard the quality of teaching and learning in intelligent education system。 .

Based on this, this study proposed a new method for factitious hallucination detection in intelligent educational system. Specifically, this study utilizes the Retrieval Augmented Generation (RAG) technique to perform knowledge base retrieval for constructing prompts based on the pre-trained ChatGLM3-6B, which enhances the model's ability to generate responses. The model is then fine-tuned using the LoRA method as a means of constructing a factitious hallucination detection model. In this paper, experiments are conducted on a real dataset in the educational domain to validate the consistency and accuracy of the model in understanding and generating answers to educational content with questions by evaluating the accuracy of the answers generated from the questions to suit the specific needs of the educational domain.

II. Related Work

A. Intelligent Education System

The development of intelligent education system has greatly facilitated the realization of personalized learning and intelligent teaching. In this field, intelligent QA technologies play a key role. Currently, QA technologies in intelligent education system are mainly categorized into three types: document set-based, knowledge graph-based and generative model-based. Document set-based Q&A system are widely used in the field of education to answer questions by analyzing a large number of documents, but still need to improve the accuracy and performance [6] .

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Knowledge graph-based QA(Question-Answer) system, on the other hand, provide highly interpretable answers by constructing a knowledge network, which helps learners to understand the connections between knowledge [7] , but face the challenges of knowledge graph construction and maintenance. Both system have their own advantages, but both need to be further optimized to better serve educational scenarios. In recent years, generative model-based QA system have been widely used in intelligent education system because they are closer to human interaction. These system are able to generate more natural and fluent answers and provide a personalized learning experience. However, most of the current generative language models are based on a large number of parameters to store knowledge, and there is a risk of over-generalization on untrained data, which is prone to generating some fictional "facts", thus triggering the phenomenon of "factual hallucination" and misleading the users[8].

B. *Factual hallucination*

In the field of Large Language Models (LLMs), factual illusion refers to the generation of seemingly correct but factually incorrect content by the model, and this problem is manifested in tasks such as text summarization, Q&A, and dialog generation, which can have adverse effects such as misinformation for critical applications. For token-level detection, HADES constructs a dataset by perturbing Wikipedia text and proposes a baseline approach based on a pre-trained model [9] , while NPH trains factual critics and optimizes dialogue generation using an external knowledge graph to detect potentially illusory entities [10] . For sentence-level detection, SelfCheckGPT [11] With the help of evaluating LLM self-consistency, which is detected by calculating consistency scores in multiple ways, ExHalder [12] Constructing frameworks and datasets for news headline hallucination, Harim+ [13] Reinterpretation of Decoder Target Definition Metrics Evaluating Summaries. HaluEval [14] proposes benchmarks and generates labeled samples to evaluate LLMs detection performance. These works provide ideas for solving the hallucination problem at different levels, but there are still challenges in detecting factual hallucination under the educational domain. The diversity and complexity of educational content and the high demand for accuracy make it difficult to directly apply traditional hallucination detection methods. Therefore, the development of specific hallucination detection techniques for the educational domain, combining pedagogical theories and advanced natural language processing methods, has become an important direction in the current research of intelligent educational system [15] .

III. Experimental Methods

The experimental methodology of this study consists of the following main steps:(1) Collect data in the educational domain and construct a knowledge vector base by semantic vectorization using Sentence-BERT. (2) Retrieve relevant information from the knowledge vector base using the RAG technique as a means of constructing a prompt optimization base model and fine-tune the model using the LoRA method to adapt to specific task requirements. (3) Construct a standard test set to comprehensively evaluate the model performance using metrics such as precision, recall and F1 score. The specific model framework is shown in Fig. 1:

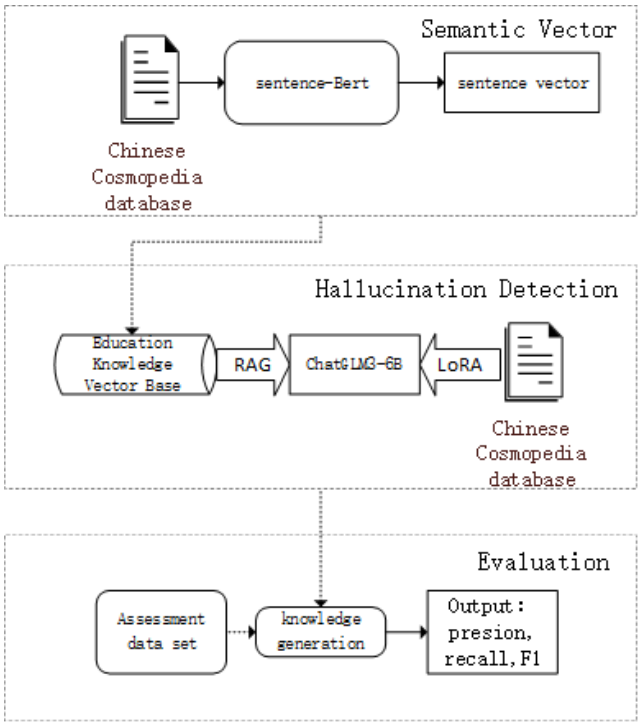


Fig. 1. Construction of the factitious hallucination detection model

A. *Knowledge Vector Library Construction*

First of all, each paragraph of text in the dataset is chunked, assuming that the text after chunking is Dij, where $i = [1,2,3,\ldots,n]$ denotes the text serial number, $j = [1,2,3,\ldots,m]$ denotes the text block serial number; then the collected data are semantically vectorized using the Sentence-BERT (SBERT) model. Specifically, each educational text D_{ij} will be converted into a fixed-dimension semantic vector c. Through the Sentence-BERT model, we construct educational knowledge vectors by mapping each educational text dij into a fixed-dimension semantic vector V_{ij} , which is denoted by SBERT as an embedding model, see Equation (1).

$$V_{ij} = \text{SBERT}(D_{ij}) \tag{1}$$

B. *Optimizing the base model*

In this study, we design a two-phase RAG architecture, including a retrieval phase and a generation phase. In the retrieval phase, given an input problem q, we first vectorize it using the sentence-BERT model to obtain the vector v_q: $v_q = SBERT(q)$. In order to improve the search efficiency of large-scale vector data, this study adopts the approximate nearest neighbor search algorithm and uses the FAISS library to construct vector indexes. The FAISS library uses a method based on inverted indexing to cluster and group similar vectors [16] . The search process is divided into two steps: first, a preliminary search is conducted among groups to roughly determine the group where the answer is located, thus narrowing the search scope and improving the retrieval efficiency [17] . Subsequently, the top k entries that are most relevant to the

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query conditions are selected from the knowledge vector base by calculating the vector similarity, and the corresponding text is returned as a reference. In the generation phase, the retrieved k entries are combined with the original question q to form an enhanced prompt. this enhanced prompt is then fed into the ChatGLM3-6B model to generate the answer.

C. Model fine-tuning

In order to further improve the performance of the model in the educational domain, this study employs the LoRA method for model fine-tuning. The core idea of LoRA is to achieve efficient model adaptation by adding small trainable rank decomposition matrices. For the original pre-trained weight matrix, LoRA introduces the introduction of a low-rank matrix $\Delta\theta$. Specifically, assuming we have a pre-trained model with parameters θ, the LoRA method updates the model parameters by introducing a low-rank tuning matrix $\Delta\theta: \theta'$.

$$\theta' = \theta + \Delta\theta \tag{2}$$

The loss function L(θ') is usually a cross-entropy loss and is used to train the generative model to output accurate answers:

$$L(\theta') = -\sum_{i=1}^{N} logP(Aq|q, Kq; \theta') \tag{3}$$

where denotes the probability that P generates the answer Aq given the question q and the associated knowledge fragment Kq. With the LoRA approach, we are able to efficiently fine-tune the model and significantly improve the performance of the model with limited computational resources.

D. Standardized Test Set Construction

In order to evaluate the model accurately, this study constructed a standardized test set. The relevant text was input into ChatGPT, and then a prompt template was used to generate the normalized Q&A, joining the question set Q and the evaluation set E as the standard test set, as shown in Fig. 2. The question set Q contains questions covering all aspects of the input text, and the evaluation set E is the reference answer corresponding to question set Q. In addition, this study used the sentence-Bert algorithm combined with cosine similarity to calculate the similarity of the generated questions, and data with similarity higher than 0.8 were merged.

IV. Experimentation

The research environment of this paper is based on Windows 11 operating system, using Python 3.10 programming language and RTX4090 with 32GB memory for algorithm experimentation. For the hyperparameter settings, the batch size is set to 16, the learning rate is set to 1e-4, the maximum length of the input sequence is set to 1536, the maximum length of the source sequence is set to 1024, and the weight decay is set to 0.1.

A. Data sets

The dataset used in this study is Chinese Cosmopedia, a large-scale Chinese synthetic dataset constructed by OpenCSG algorithm team, which contains about 15 million pieces of data and about 60B tokens, and covers platforms such as Chinese Wikipedia, Baidu Encyclopedia, Zhihu Q&A, and technical blogs, to ensure the breadth and authority of the content. Covering a variety of different styles such as university textbooks, secondary school textbooks, stories for young children, general stories, and wikihow style tutorials, the content is well structured, exploring the core concepts of various disciplines in depth, or concise and easy to understand, focusing on the communication of basic knowledge.

By preprocessing the Chinese Cosmopedia dataset, 100,000 pieces of data were finally selected for knowledge vector base construction. 10,000 data are selected for LoRA to fine-tune the model with a ratio of 8:1:1 for the training, validation and test sets. 1,000 data are randomly selected for the construction of the standard test set.

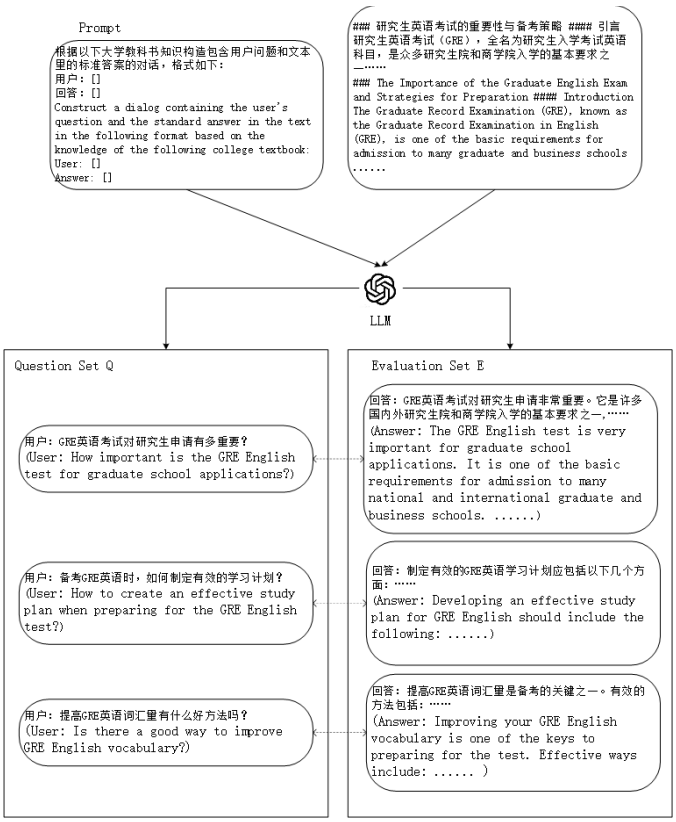


Fig. 2. Standardized test set construction diagram

B. Evaluation indicators

In this paper, three metrics, Precision, Recall, and F1 Score, are used to evaluate the model. They are used to measure how well the model performs under different problem categories. Precision is the ratio of the number of samples correctly predicted by the model to the total number of samples, Recall is the ratio of the number of positive samples correctly predicted by the model to the actual number of positive samples, and the F1 Score is the reconciled average of Precision and Recall, which are all in the range of 0 to 1, with the higher value representing better model performance. The specific formulas for these three measures are as follows.

$$\text{Recall} = \frac{\text{TP}}{\text{TP+FN}} \tag{4}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP+FP}} \tag{5}$$

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$$\text{F1 Score} = \frac{2*P*\text{Recall}}{P+\text{Recall}} \quad (6)$$

Where TP represents the number of model-generated answers that match the reference answer. FN represents the number of modeled answers that are present in the reference answer but not in the model-generated answers. FP is the number of model-generated answers in this study that are not present in the reference answer.

C. Experimental results and analysis

a) Analysis of model effects: This study aims to compare the performance of our proposed new model with the existing ChatGLM3-6B model in processing different types of educational texts. This is done by testing five different types of texts including university textbooks, secondary school textbooks, early childhood textbooks, general stories and wikihow. Using the ChatGLM3-6B model through the Zero-shot approach, questions are directly fed into the model and there the model directly generates the answers based on its knowledge inference ability gained from training. In this study, the standard test set was used for the evaluation, and the questions in the standard test set were compared with the reference answers by generating answers through this study's model and ChatGLM, respectively, and then a series of data on the model's accuracy, recall, and F1 value were obtained by calculation, as shown in Table I.

TABLE I. ANSWER GENERATION EVALUATION FORM

Type	Our Model			ChatGLM3-8B		
	Precision	***Recall***	***F1***	***Precision***	***Recall***	***F1***
copy	0.79	0.78	0.79	0.75	0.76	0.76
Secondary school	0.84	0.86	0.85	0.83	0.81	0.82
Early childhood	0.88	0.90	0.87	0.86	0.85	0.87
Common	0.83	0.85	0.84	0.80	0.82	0.83
wikihow	0.80	0.83	0.81	0.77	0.77	0.78

As can be seen from Table I, the present study model performed better on several metrics relative to the ChatGLM3-6B model. The fact that the present study model performs well on these metrics indicates that it has better answering ability than the ChatGLM3-6B model on various types of questions. This may be due to the fact that the model of this study incorporates the RAG technique for knowledge retrieval and fine-tuning using LoRA during the construction process, and the model is able to better extract and process the key information in the questions, thus generating answers more accurately.

In addition, the data in the table shows that the model performs relatively weakly on the secondary school textbook and university textbook category categories. Upon analysis, this may be due to reflecting the fact that more complex concepts and more specialized terminology in advanced educational content increase the risk of factual illusions.

b) Examples of model responses:

This study further qualitatively discusses the model generation content by selecting representative examples of problems for analysis. Where the red labeled sections indicate that they match the reference answer, the underlined sections are the model reasoning sections, and the blue labeled sections are the obvious error sections. As shown in Fig. 3.

As can be seen from Fig. 3, the results generated by the model of this study show significant advantages. First, the model of this study can perfectly match all the key points in the reference answers and is more in line with human expression habits in terms of language expression, in addition to which the model demonstrates an in-depth understanding of the question and the ability to analyze it. In contrast, the responses of the ChatGLM3-8B model cover some basic information but fail to fully cover all the key points in the reference answers. At the same time, the generated responses contained inaccurate information, such as incorrectly mentioning that the GRE contains a "language section". This kind of error is typical of factual hallucination, which may mislead the user's cognition and further reduce the reliability of the model. In this study, the model was fine-tuned using the RAG technique combined with the knowledge vector base retrieval and the LoRA method to lighten the model, and the hallucination phenomenon was reduced compared to the comparison model. These advantages make the model valuable in enhancing the intelligence, accuracy, and interactivity of intelligent education system.

Question	Model for this study	ChatGLM3-6B	Reference Answer
How important is the GRE English test to graduate school applications?	①The GRE English test plays a crucial role in the graduate school application process. First and foremost,② it is required for admission to many of the top graduate and business schools, both nationally and internationally. A good score on the GRE not only demonstrates your English proficiency, but also ③your ability to pursue graduate studies. Second, the GRE helps test takers identify and improve their language learning weaknesses by comprehensively evaluating ④the four areas of listening, speaking, reading, and writing. Finally, the test also focuses on ⑤developing candidates' logical analysis and critical thinking skills, laying a solid foundation for future academic research. Therefore, the GRE is not only a gateway to admission, but also an important way to improve overall academic ability.	①The importance of the GRE English test (especially the language portion of it) for graduate school applications is as follows: Threshold for Admission: Many graduate programs use GRE scores as one of the basic requirements for application, and a low score may directly lead to rejection of the application. Demonstration of competitiveness: A high GRE language score can help an applicant stand out from the competition by ⑤demonstrating critical thinking, reading comprehension, and language use skills. Scholarship evaluation: GRE scores are often an important factor in scholarship evaluations, and a high score may increase the chances of receiving a scholarship. Background Complement: For applicants who are not native English speakers, GRE English scores can demonstrate English proficiency and help make up for deficiencies in language background.	①The GRE English test is very important for graduate school applications.② It is one of the basic requirements for admission to many national and international graduate and business schools. A high score not only proves your English proficiency, ③but also demonstrates that you have what it takes to pursue a graduate program. In addition, the GRE exam comprehensively examines a candidate's ability to utilize language in a comprehensive manner, including ④listening, speaking, reading, and writing, which helps to identify weaknesses in language learning. It also ⑤develops candidates' logical and critical thinking skills, preparing them for future academic studies.

Fig. 3. Example analysis diagram

V. SUMMARY AND OUTLOOK

In this paper, a RAG-based factitious illusion detection model is proposed, using RAG technique to retrieve the knowledge vector base, optimizing the prompt of the base model, and then fine-tuning the model through the LoRA method to improve the adaptability of the model in the educational domain.

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Finally, a standard test set is constructed and the model performance is comprehensively evaluated using metrics such as accuracy, recall, and F1. Meanwhile, there is still room for improvement in the methodology of this paper, which is summarized as follows:1) The knowledge vector base used in the current model may not be able to reflect the rapidly updated educational content in a timely manner. A dynamic knowledge updating mechanism needs to be explored to ensure that the model can adapt to the ever-changing knowledge in the educational domain.2): the retrieval efficiency of the RAG technique may become a bottleneck on large-scale educational datasets. Optimizing the retrieval algorithm or adopting more advanced information retrieval techniques, such as dense retrieval or hybrid retrieval methods, can be considered in the future.3) Currently, the model mainly handles textual inputs, but the educational content often contains multimodal information, such as images and videos. Future research can explore how to effectively integrate multimodal inputs to improve the comprehensiveness of detection.

ACKNOWLEDGMENT

This work was supported in part by National Natural Scien ce Foundation of China (Grant No. 62377020) named "Researc h on the Intelligent Evaluation Method of Teacher Information Technology Application Ability based on Classroom Teaching Behavior Characteristics", Fundamental Research Funds for th e Central Universities named "Research on Intelligent Evaluati on Technology and Application of Teachers' Teaching Ability based on Multimodal Data" (CCNU22JC011), Open Fund of H ubei Key Laboratory of Digital Education (Grant No. F2024K0 2) named "Research on the Key Technology of Teachers' Digit al Teaching Competence Assessment Based on the Characterist ics of Teaching Behavior in the Classroom".

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